

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{3} m_2 \tilde{\mu} g_2^4 y_d^{i1i2} LF_{2,2,0}[m_2, \tilde{\mu}] + \right. \\
& \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_d^{i1i2} LF_{3,2,-1}[m_2, \tilde{\mu}] + \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d^{pr}} LF_{2,2,0}[m_d^r, m_q^p] + \\
& \frac{1}{4} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e^{pr}} LF_{2,2,0}[m_e^r, m_l^p] - \frac{1}{2} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e^{pr}} LF_{3,1,0}[m_l^p, m_e^r] - \\
& \frac{1}{4} \tilde{\mu} g_1^2 y_d^{i1i2} y_e^{pr} \overline{a_e^{pr}} LF_{3,2,-1}[m_l^p, m_e^r] + \frac{3}{8} \tilde{\mu} y_d^{i1i2} y_e^{pr} \overline{a_e^{pr}} (3 g_1^2 + g_2^2) LF_{4,1,-1}[m_l^p, m_e^r] - \\
& \frac{1}{6} \tilde{\mu} y_d^{i1i2} y_e^{pr} \overline{a_e^{pr}} (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[m_l^p, m_e^r] - \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d^{pr}} LF_{3,1,0}[m_q^p, m_d^r] - \frac{1}{4} \tilde{\mu} g_1^2 y_d^{pr} y_d^{i1i2} \overline{a_d^{pr}} LF_{3,2,-1}[m_q^p, m_d^r] + \\
& \frac{3}{8} \tilde{\mu} y_d^{pr} y_d^{i1i2} \overline{a_d^{pr}} (5 g_1^2 + 3 g_2^2) LF_{4,1,-1}[m_q^p, m_d^r] - \\
& \frac{1}{2} \tilde{\mu} y_d^{pr} y_d^{i1i2} \overline{a_d^{pr}} (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[m_q^p, m_d^r] + \\
& \frac{1}{8} \tilde{\mu} y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} (g_1^2 - 5 g_2^2) LF_{2,2,0}[m_q^p, m_u^r] - \frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} LF_{3,1,0}[m_q^p, m_u^r] + \\
& \frac{1}{2} \tilde{\mu} g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} LF_{3,2,-1}[m_q^p, m_u^r] + \frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} LF_{4,1,-1}[m_q^p, m_u^r] - \\
& \frac{1}{4} \tilde{\mu} y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} (7 g_1^2 + 2 g_2^2) LF_{3,1,0}[m_u^r, m_q^p] - \\
& \frac{1}{8} \tilde{\mu} y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} (g_1^2 - 5 g_2^2) LF_{3,2,-1}[m_u^r, m_q^p] + \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} (5 g_1^2 + 2 g_2^2) LF_{4,1,-1}[m_u^r, m_q^p] - \frac{1}{2} \tilde{\mu} y_d^{i1i2} y_u^{pr} \overline{a_u^{pr}} \\
& (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[m_u^r, m_q^p] - \frac{1}{4} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{3,1,0}[\tilde{\mu}, m_1] + \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_1] + \frac{1}{8} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (6 g_1^2 + 5 g_2^2) LF_{4,1,-1}[\tilde{\mu}, m_1] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_1] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{1}{12} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (9 g_1^2 + 5 g_2^2) LF_{3,1,0}[\tilde{\mu}, m_2] + \frac{7}{3} m_2 \tilde{\mu} g_2^4 y_d^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{1}{8} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (18 g_1^2 + 11 g_2^2) LF_{4,1,-1}[\tilde{\mu}, m_2] - \\
& 2 m_2 \tilde{\mu} g_2^4 y_d^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_2] - \frac{1}{2} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (3 g_1^2 + 2 g_2^2) LF_{5,1,-2}[\tilde{\mu}, m_2] - \\
& \frac{1}{144} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, m_d^{i2}, m_q^{i1}] - \\
& \frac{1}{144} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, m_q^{i1}, m_d^{i2}] + \\
& \frac{1}{24} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (-7 g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_d^{i2}] + \\
& \frac{1}{48} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (-7 g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (7 m_1 + 5 m_2) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{3}{16} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (g_1^2 - 7 g_2^2) \\
& LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^{i1}] + \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_3, m_d^{i2}, m_q^{i1}] + \\
& \frac{1}{6} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_3, m_q^{i1}, m_d^{i2}] - \frac{1}{216} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - 3 g_2^2) \\
& LF_{2,1,1,0}[m_d^{i2}, m_1, m_q^{i1}] - \frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_d^{i2}, m_1, m_q^{i1}] - \\
& \frac{1}{18} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} LF_{2,1,1,0}[m_d^{i2}, m_1, \tilde{\mu}] + \frac{1}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 - 3 g_2^2) \\
& LF_{2,1,1,0}[m_d^{i2}, m_3, m_q^{i1}] + \frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_d^{i2}, m_3, m_q^{i1}] - \\
& \frac{3}{4} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} \overline{a_d^{pr}} LF_{2,2,1,-1}[m_q^p, m_q^s, m_d^r] + \frac{3}{4} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} \\
& LF_{2,1,1,0}[m_q^p, m_q^s, m_u^r] - \frac{3}{4} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{3,1,1,-1}[m_q^p, m_q^s, m_u^r] - \\
& \frac{9}{8} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{2,2,1,-1}[m_q^p, m_u^r, m_q^s] - \frac{1}{24} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (g_1^2 + 9 g_2^2) \\
& LF_{2,1,1,0}[m_q^p, m_u^r, \tilde{\mu}] + \frac{1}{8} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_q^p, m_u^r, \tilde{\mu}] + \\
& \frac{1}{16} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_q^p, \tilde{\mu}, m_u^r] + \frac{3}{4} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} \\
& LF_{2,1,1,0}[m_q^s, m_q^p, m_u^r] - \frac{3}{4} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{3,1,1,-1}[m_q^s, m_q^p, m_u^r] + \\
& \frac{3}{8} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{2,2,1,-1}[m_q^s, m_u^r, m_q^p] + \frac{1}{216} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 - 3 g_2^2) \\
& LF_{2,1,1,0}[m_q^{i1}, m_1, m_d^{i2}] - \frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_q^{i1}, m_1, m_d^{i2}] - \\
& \frac{1}{72} m_1 \tilde{\mu} g_1^2 y_d^{i1i2} (g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_q^{i1}, m_1, \tilde{\mu}] + \frac{1}{8} m_2 \tilde{\mu} g_2^2 y_d^{i1i2} (-g_1^2 + g_2^2) \\
& LF_{2,1,1,0}[m_q^{i1}, m_2, \tilde{\mu}] - \frac{1}{9} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 - 3 g_2^2) LF_{2,1,1,0}[m_q^{i1}, m_3, m_d^{i2}] + \\
& \frac{1}{3} m_3 \tilde{\mu} g_3^2 y_d^{i1i2} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_q^{i1}, m_3, m_d^{i2}] + \frac{3}{2} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} \\
& LF_{2,1,1,0}[m_u^r, m_q^p, m_q^s] - \frac{3}{2} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{3,1,1,-1}[m_u^r, m_q^p, m_q^s] - \\
& \frac{1}{24} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (11 g_1^2 + 3 g_2^2) LF_{2,1,1,0}[m_u^r, m_q^p, \tilde{\mu}] + \\
& \frac{1}{8} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_u^r, m_q^p, \tilde{\mu}] - \\
& \frac{3}{2} \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{2,2,1,-1}[m_u^r, m_u^t, m_q^p] + \\
& \frac{1}{16} \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_u^r, \tilde{\mu}, m_q^p] + \\
& 3 \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{2,1,1,0}[m_u^t, m_q^p, m_u^r] - \\
& 3 \tilde{\mu} y_d^{i1i2} \overline{y_u^{st}} y_u^{pt} y_u^{sr} \overline{a_u^{pr}} LF_{3,1,1,-1}[m_u^t, m_q^p, m_u^r] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + \frac{1}{8} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (9 m_1 + 5 m_2) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] - \\
& \frac{3}{4} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \\
& \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \frac{1}{9} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \\
& LF_{2,1,1,1,-1}[m_1, m_d^{i2}, m_q^{i1}, \tilde{\mu}] - m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{y_u^{pr}} y_u^{i1r} LF_{1,1,1,1,0}[m_2, m_q^{i1}, m_q^p, \tilde{\mu}] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d^{pr}} LF_{2,2,1,1,-1}[m_d^r, m_q^p, m_d^t, m_q^s] - \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d^{pr}} LF_{2,2,1,1,-1}[m_d^t, m_q^p, m_d^r, m_q^s] + \\
& \frac{1}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \overline{a_e^{pr}} LF_{2,2,1,1,-1}[m_e^r, m_l^p, m_e^t, m_l^s] - \\
& \frac{1}{4} \tilde{\mu}^3 y_d^{i1i2} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \overline{a_e^{pr}} LF_{2,2,1,1,-1}[m_e^t, m_l^p, m_e^r, m_l^s] + \\
& \frac{1}{2} \tilde{\mu}^3 y_d^{i1i2} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \overline{a_e^{pr}} LF_{2,1,1,1,0}[m_l^p, m_e^r, m_e^t, m_l^s] - \\
& \frac{1}{2} \tilde{\mu}^3 y_d^{i1i2} \overline{y_e^{st}} y_e^{pt} y_e^{sr} \overline{a_e^{pr}} LF_{3,1,1,1,-1}[m_l^p, m_e^r, m_e^t, m_l^s] + \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d^{pr}} LF_{2,1,1,1,0}[m_q^p, m_d^r, m_d^t, m_q^s] - \\
& \frac{3}{2} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{pt} y_d^{sr} y_d^{i1i2} \overline{a_d^{pr}} LF_{3,1,1,1,-1}[m_q^p, m_d^r, m_d^t, m_q^s] - \\
& \frac{3}{4} \tilde{\mu} y_d^{sr} y_d^{i1i2} \overline{a_d^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^r, m_u^t] - \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d^{st}} y_d^{pt} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} LF_{2,2,1,1,-1}[m_q^p, m_q^s, m_d^t, m_u^r] + \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t] - \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{3,1,1,1,-1}[m_q^p, m_q^s, m_u^r, m_u^t] + \\
& \tilde{\mu} y_d^{si2} \overline{y_u^{st}} y_u^{pt} y_u^{i1r} \overline{a_u^{pr}} LF_{1,1,1,1,0}[m_q^p, m_q^s, m_u^r, \tilde{\mu}] - \\
& \frac{9}{8} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_q^p, m_u^r, m_q^s, m_u^t] + \\
& \tilde{\mu} y_d^{pi2} \overline{y_u^{st}} y_u^{sr} y_u^{i1t} \overline{a_u^{pr}} LF_{1,1,1,1,0}[m_q^p, m_u^r, m_u^t, \tilde{\mu}] - \\
& \frac{3}{8} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_q^p, m_u^t, m_q^s, m_u^r] + \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,1,1,1,0}[m_q^s, m_q^p, m_u^r, m_u^t] - \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{3,1,1,1,-1}[m_q^s, m_q^p, m_u^r, m_u^t] + \\
& \frac{3}{8} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_q^s, m_u^r, m_q^p, m_u^t] - \\
& \frac{3}{8} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_q^s, m_u^t, m_q^p, m_u^r] + \\
& \frac{3}{2} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,1,1,1,0}[m_u^r, m_q^p, m_q^s, m_u^t] - \\
& \frac{3}{2} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{3,1,1,1,-1}[m_u^r, m_q^p, m_q^s, m_u^t] - \\
& \frac{3}{4} \tilde{\mu} y_d^{i1i2} y_u^{sr} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{2,2,1,1,-1}[m_u^r, m_u^t, m_q^p, m_q^s] - \\
& \frac{1}{6} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_d^{i2}] - \\
& \frac{1}{3} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_q^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,0}[\tilde{\mu}, m_1, m_d^r, m_q^p] - \\
& \frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{2,1,1,1,0}[\tilde{\mu}, m_2, m_d^r, m_q^p] + \\
& \tilde{\mu} g_2^2 y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_q^p, m_u^r] - \\
& \frac{1}{6} \tilde{\mu} g_1^2 g_2^2 y_d^{i1i2} (m_1 + m_2) LF_{1,1,1,1,1,-1}[m_1, m_2, m_d^{i2}, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{6} m_1 g_1^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,1,0}[m_1, m_d^r, m_q^{i1}, m_q^p, \tilde{\mu}] - \\
& \frac{1}{3} m_1 g_1^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,1,0}[m_1, m_d^{i2}, m_d^r, m_q^p, \tilde{\mu}] - \\
& \frac{1}{9} m_1 g_1^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,1,0}[m_1, m_d^{i2}, m_d^r, m_q^{i1}, m_q^p] - \\
& \frac{1}{2} m_2 g_2^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,1,0}[m_2, m_d^r, m_q^{i1}, m_q^p, \tilde{\mu}] - \\
& m_2 \tilde{\mu} g_2^2 y_d^{pi2} \overline{a_u^{pr}} a_u^{i1r} LF_{1,1,1,1,1,0}[m_2, m_q^{i1}, m_q^p, m_u^r, \tilde{\mu}] + \\
& \frac{8}{3} m_3 g_3^2 \tilde{\mu}^3 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} LF_{1,1,1,1,1,0}[m_3, m_d^{i2}, m_d^r, m_q^{i1}, m_q^p] + \\
& \left. \tilde{\mu} y_d^{si2} y_u^{i1r} \overline{a_u^{pr}} \overline{a_u^{st}} a_u^{pt} LF_{1,1,1,1,1,0}[m_q^p, m_q^s, m_u^r, m_u^t, \tilde{\mu}] \right)
\end{aligned}$$